TSINGHUA SCIENCE AND TECHNOLOGY
ISSN 1007-0214 22/37 pp329–343
DOI: 10.26599/TST.2025.9010067
Volume 32, Number 1, February 2027

Research Article

Multi-Agent Reinforcement Learning Optimization for Virtual Coupled Train Set Problem in CBTC Systems

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Abstract: Modern urban rail transit systems face unprecedented challenges and opportunities due to accelerated urbanization and growing travel demand, which increase the need for operational efficiency and service quality. Traditional approaches based on Communication-Based Train Control (CBTC) systems are insufficient for fluctuating passenger flows. This study introduces a Virtual Coupled Train Set (VCTS) controlled by Multi-Agent Reinforcement Learning (MARL) to address these issues. VCTS enhances capacity and efficiency by dynamically adjusting train spacing and formations through train-to-train communication and onboard sensors. The MARL framework optimizes collaborative control in high-dimensional state spaces, complex action spaces, and dynamic environments. Simulation results demonstrate that the proposed MARL algorithm significantly improves train operation efficiency and safety compared to both traditional methods and previous reinforcement learning approaches. This research advances the application of the Internet of Things in rail transit, offering innovative solutions for safer, more efficient, and intelligent urban rail systems.

Key words: reinforcement learning; Virtually Coupled Train Sets (VCTSs); Communication-Based Train Control (CBTC)

1 Introduction

The modern urban rail transit system is currently experiencing both unique challenges and opportunities due to factors such as advances in technology, increasing demand for sustainable transportation, and the need for infrastructure upgrades. On the one hand, the accelerated process of urbanization and the growing travel demand of residents have put higher demands on the operational efficiency and service quality of rail transit; on the other hand, the development and application of emerging technologies provide new ideas and possibilities for solving these problems.

Traditional urban rail transit adopts a Communication-Based Train Control (CBTC) system[1], which realizes automatic driving and dispatching of trains through train-to-ground communication and automatic train protection, ensuring driving safety. This model typically uses a uniform train departure interval, i.e., trains depart

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Manuscript received: 2024-07-18; revised: 2024-11-17; accepted: 2025-04-10

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according to a fixed schedule[2]. However, with the continuous growth of passenger flow and the uneven distribution of passenger flow (such as during peak hours and sudden passenger flows during large events), the existing train operation plans are increasingly unable to meet actual demand, and the contradiction between train capacity and passenger flow demand is becoming more prominent.

To further improve the operational efficiency and service quality of the rail transit system, it is urgent to introduce innovative operational organization methods and advanced technical means. Among them, the Virtual Coupled Train Set (VCTS)[3] is a new type of train formation and operation control method that can better adapt to passenger flow fluctuations and improve line transportation capacity by dynamically adjusting train spacing and flexibly changing train formations.

VCTS uses collaborative control between trains using using Train-to-Train (T2T) and train-to-ground communication along with onboard sensors. Instead of physical coupling, VCTS trains use reliable wireless tech to maintain an "electronic coupling", allowing dynamic adjustment of train spacing based on operational needs. Combined with onboard lidar, millimeter-wave radar, vision sensors, and other equipment, VCTS can support high automation and intelligence of train operations while ensuring driving safety.

Trains of VCTS can achieve autonomous recoupling and decoupling operations based on transport demand and planning T2T wireless communication[4]. In the VCTS system based on the Industrial Internet of Things (IIoT), to successfully achieve the control objectives of headway and speed, all other units must be coordinated through the Train Cooperation Controller (TCO), where the TCO, as the core of the control strategy, is responsible for maintaining the uniform linear formation of the leading train and the continuous operation path of the overall train by adjusting the speed and headway of the following trains[5]. Furthermore, in such a VCTS system, the concept of braking distance has also changed from absolute braking distance to relative braking distance. This shift means that the implementation of braking measures is more reliant on the relative position between trains and the dynamic environment[6].

Such an urban rail transit system based on VCTS can significantly enhance line capacity and operational efficiency, precisely match passenger flow with transport capacity, and provide passengers with more comfortable and convenient travel services. At the same time, as a new train operation control method, VCTS also puts higher demands on the safety, reliability, and intelligence level of rail transit, necessitating systematic research and exploration in key technology development, testing, validation, and standardization. Therefore, the member trains of VCTS have a certain degree of autonomy and require strict timeliness and reliability in wireless communication. However, in the VCTS scenario, the complexity of the communication network and the high-speed mobility of trains pose severe challenges to achieving optimal collaborative control. Traditional strategies cannot meet the performance requirements of VCTS. Multi-Agent Reinforcement Learning (MARL)[7–11] allows trains to intelligently explore optimal solutions[12] in this complex environment. Thus, reinforcement learning control has become an emerging technology in optimizing VCTS collaborative control within CBTC systems. Specifically, the new generation of IIoT[13, 14] connects a large number of user devices and sensors to local units (such as edge devices in edge computing) with some basic storage and computing resources through high-speed 5th/6th Generation Mobile Communication Technology (5G/6G) networks. Therefore, a series of VCTS applications can be deployed on train carriages close to data resources, providing users with real-time and intelligent services. Fraga-Lamas[15] proposed using IIoT to improve railway safety levels. IIoT is also used to enhance railway, including track and facility maintenance capabilities[5]. With the integration of IIoT, big data, and artificial intelligence, railway maintenance becomes more efficient and intelligent, enabling condition-based maintenance[16]. Similarly, trains can move synchronously like VCTS, meaning multiple trains can be regarded as a fleet, thereby increasing train capacity.

However, there is little research on the performance evaluation and analysis of IIoT-based VCTS, which is urgently needed to support the application of IIoT in Urban Rail Transit (URT). Due to the dynamic characteristics of trains and the critical requirements for T2T communication, achieving VCTS consistency and coverage control is challenging. The number of IIoT states, including numerous train devices and sensing nodes, will be so large that no centralized machine learning algorithm can be executed. In

addition, exploring machine learning in an overly large state space encounters the following challenges:

(1) If small exploration steps are used in an overly large state space, exploration efficiency will be severely insufficient, and convergence speed will be slow or may not converge.

(2) When exploration steps are too large, a large amount of space will not be fully explored, making it easy to miss the optimal solution or cause severe oscillations.

To overcome the challenges of overly large state spaces in multi-agent machine learning and the limited exploration capabilities of current multi-agent ML technologies, this paper proposes not to let multi-agent ML algorithms randomly choose a state as the initial state but to judge a good state as the starting point for multi-agent ML. A good state means that the state of the nodes in multi-agent ML is not far from the optimal solution; the exploration space has been limited. Based on the relative information between trains, VCTS can make optimized decisions, ensuring the stable and safe operation of the train system and achieving precise matching of passenger flow and transport capacity.

This paper proposes using MARL to solve the optimal collaborative control strategy of VCTS. The designed MARL framework addresses the high-dimensional state space, complex action space, dynamic environmental changes, and multi-agent coordination in VCTS scenarios through key technologies such as state compression, action space search engine optimization, dynamic environment adaptation, heuristic methods, and multi-agent collaborative learning, achieving effective collaborative control of VCTS under the premise of meeting safety constraints. Simulation experiments show that the proposed MARL algorithm has better effects in improving train operation efficiency and safety compared to traditional methods. By combining MARL with VCTS and innovatively addressing the challenges of complexity and mobility, this paper helps advance train collaborative control in large-scale CBTC systems and VCTS, ultimately improving the efficiency and safety of railway transportation.

The main contributions of this paper are

(1) Proposes an IIoT-based VCTS collaborative control framework, utilizing vehicle-to-vehicle communication and onboard sensors to achieve collaborative control between trains, improving transportation efficiency and safety in rail transit.

(2) Designs a multi-agent reinforcement learning algorithm that addresses the high-dimensional state space, complex action space, dynamic environments, and multi-agent coordination in VCTS scenarios through key technologies such as state compression, action space search engine optimization, dynamic environment adaptation, heuristic methods, and multi-agent collaborative learning.

(3) Validates the effectiveness of the proposed method through simulation experiments, showing that the proposed MARL algorithm has better effects in improving train operation efficiency and safety compared to traditional methods.

The research results of this paper are of great significance for promoting the application of IIoT in the field of rail transit, providing new ideas and methods for achieving safer, more efficient, and intelligent rail transit systems. Future work will further optimize the algorithms, improve their convergence speed and stability, and test and apply them in actual rail transit systems. Additionally, we explore extending the methods proposed in this paper to other IIoT application scenarios, such as autonomous driving and smart cities, to achieve broader applications.

The remainder of the paper is organized as follows. Section 2 provides the related work. Section 3 describes the system model and problem formulation. Section 4 presents the description of heuristically accelerated MARL. Section 5 describes our implementation and shows simulation results. We conclude this paper in Section 6.

2 Related Work

A typical CBTC system is essential for modern urban rail transit, ensuring safe and efficient operations through its structured components[1, 17, 18]. At the heart of the system is the ground control center[19], which includes key elements such as Automatic Train Supervision (ATS), Zone Controller (ZC), and Computer Interlocking (CI). These components work together to oversee and manage the overall system operations. The Data Storage Unit (DSU) within the ground control center is responsible for storing all necessary data, facilitating continuous and reliable system performance. The train subsystem plays a critical role by collecting various line data and generating operation commands based on instructions received from the ground control center. This subsystem is equipped with the Vehicle On-Board

Controller (VOBC), which contains both the Automatic Train Protection (ATP) and Automatic Train Operation (ATO) modules. These modules ensure that trains operate within safe speed limits and follow the Movement Authority (MA) directives generated by the ZC.

Communication between the ground control center and the train subsystem is maintained through the Data Communication System (DCS). The DCS includes Base Band Units (BBU), Remote Radio Units (RRU), and a backbone network, which together establish robust communication links[20]. Trains continuously send packets containing critical information such as speed, position, and state to the ZC at each communication cycle moment. The ZC then uses this information, alongside track occupancy data provided by track-side equipment and routes arranged by the CI, to calculate and issue MAs[21].

The MA determines the safe distance to the nearest obstacle, including other trains or track switches, ensuring safe train operations. High-precision control algorithms enhance the efficiency and safety of the system by providing accurate MAs. The ATO subsystem[22] on the train uses the MA to generate commands for accelerating, cruising, or braking, thereby maintaining optimal train performance.

2.1 IIoT CBTC-based VCTS

Sudden surges in passenger flow, such as those caused by concerts, sports matches, and holidays, present a significant challenge for URT systems. These events can create unexpected spikes in demand, requiring more trains in one direction while the opposite direction needs fewer trains[23].

VCTS is designed to address this issue. Unlike traditional CBTC systems, where trains run at fixed intervals, VCTS allows trains to operate closer together, forming a train platoon. This setup increases capacity as trains can maintain smaller intervals among one another[24]. In a VCTS, trains in the high-demand direction stop at each station to pick up passengers, while in the low-demand direction, only the first train in the platoon makes stops, thus optimizing train use and reducing listed capacity[23].

The IIoT enhances CBTC systems by enabling real-time data collection and analysis. IIoT devices gather information from train and track components, which is processed using advanced analytics and machine learning to predict and prevent issues[25].

Safety and reliability also improve with IIoT. Real-time data allows for quick and accurate decision-making. Additionally, IIoT helps optimize energy use by adjusting train operations based on current track conditions and traffic flow, leading to significant energy savings[26]. In a VCTS, mechanical couplers are replaced with wireless communication, enabling trains to operate as a cohesive unit. The system leverages real-time data exchange to maintain safe distances between trains, thereby enhancing line capacity and operational flexibility[27]. For both URT and High-Speed Rail (HSR), VCTS can handle emergency situations and sudden passenger surges. The proposed model takes into account the effects of high-speed train movements and potential interferences, making it adaptable for both types of rail systems. Thus, with appropriate configurations, VCTS can significantly enhance rail control systems for various scenarios[28].

2.2 MARL for VCTS

Several studies have extensively examined the efficacy of real-time learning through environmental data acquisition[8, 10, 12, 29, 30]. This method has proven to enhance the performance of algorithms, particularly in the context of MARL. In recent years, MARL has gained significant interest due to its innovative approach to solving complex problems. MARL involves scenarios where two or more independent agents interact with the environment and each other. These agents' actions are influenced by their surroundings, and they may exhibit varying degrees of cooperation or competition. Consequently, MARL has found applications in diverse areas such as gaming, robotics, communication, and transportation systems.

One promising application of MARL is in optimizing VCTS. In a VCTS system, each train unit acts as an independent agent within an MARL framework. These agents (train units) interact with the shared environment (railway network) and with each other to optimize their collective performance[5]. For instance, they need to synchronize speeds, acceleration, and braking to maintain optimal distances, ensure safety, and minimize energy consumption. In multi-agent scenarios, agents face numerous choices for their next actions. If we consider all agents and their potential actions as a single entity, the size of this space increases exponentially with the number of agents. However, an extensive action space can hinder the algorithm's resolution process, affecting the scalability of the agent count[31].

In the context of VCTS, managing this vast action space is crucial. Effective MARL algorithms can handle this complexity by learning optimal policies that enable the train sets to operate efficiently as a virtually coupled unit. This involves coping with dynamic environmental changes, such as varying passenger loads and track conditions, and making real-time decisions that balance safety, efficiency, and energy consumption.

3 System Model and Problem Formulation

We introduce the model of the VCTS and the problems that it confronts. Firstly, we provide a simplistic scenario description, followed by an elaborate mathematical model of the VCTS in Section 3.1, including the train dynamics model and artificial potential field functions. In Section 3.2, the core problem that this research aims to solve is outlined.

A VCTS consists of a group of trains that operate according to a preordained plan, with the stipulation that each train must halt at every station. The schematic representation of the concept is illustrated in the corresponding Fig. 1. Therefore, the operational objectives of a VCTS are subject to temporal changes. In the broader sense, a VCTS can be perceived as a multi-agent system, and its operations can be modeled as a motion planning problem. However, due to the complex and dynamic operational conditions, achieving global information awareness and swiftly arriving at optimal decisions is challenging.

3.1 Train motion model

When studying train control systems, it is vitally important to establish a motion model of the train. This motion model can help us analyze the train's motion characteristics, predict the train's motion state, and design suitable control strategies. Typically, the dynamic model of the train can be modeled as the following discrete-time invariant linear system:

$$\begin{cases} x[k+1] = Ax[k] + Bu[k] + C\eta[k], \\ z[k] = C_1 x[k] + D_1 u[k], \\ y[k] = C_2 x[k]. \end{cases}$$

where $x[k]$ is the train state at the k-th communication cycle moment, $\eta[k]$ is the noise at the k-th communication cycle moment, $z[k]$ is the rated output at the k-th communication cycle moment, $y[k]$ is the measure of the system state, and $u[k]$ is the system input at the k-th communication cycle moment. A, B, C, C_1, C_2, and D_1 are parameter matrices. This discrete-time invariant linear system describes the variation rule of the train state in each communication cycle moment as well as the relationship between the train state and system input, output, and noise. In this discrete-time invariant linear system, the train state $x[k]$ changes with each communication cycle moment. These changes stem from three components: Firstly, the component from the previous communication cycle moment train $x[k]$, multiplied by the state transition matrix A; secondly, based on the previous communication cycle system input $u[k]$, multiplied by the input matrix B; thirdly, based on the previous communication moment noise $\eta[k]$, multiplied by the noise matrix C. The sum of these three components results in the new train state $x[k+1]$. The rated output of the train $z[k]$ comprises two parts: Firstly, the train state $x[k]$ multiplied by the output matrix C_1; secondly, the system input $u[k]$ multiplied by the direct item matrix D_1. The sum of these two parts results in the rated output of the train $z[k]$.

The measure of the train state $y[k]$ is the train state $x[k]$ multiplied by the measure matrix C_2. This reflects our observation of the train state. The dynamic model defines the train state as follows:

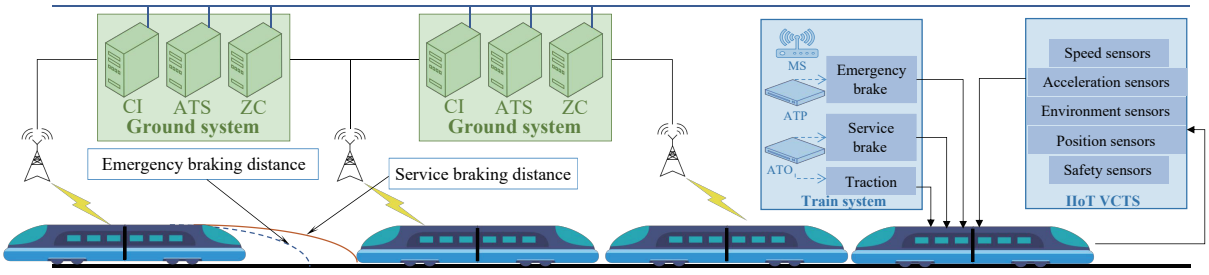


Fig. 1 CBTC system is a signaling system that uses telecommunications between the train and track equipment for traffic management and control. VCTS is a component within CBTC that integrates speed, acceleration, environment, position, and safety sensors to monitor and control train operations. CBTC-based VCTS combines these technologies to enhance train safety, efficiency, and reliability by leveraging advanced sensor data and real-time communication.

$$x[k] = [s_i[k],\ v_i[k],\ a_i[k]]^{\top},$$

where $s_i[k]$ and $v_i[k]$ represent the position and velocity of the i-th train at the k-th communication cycle moment respectively, and $\mathrm{Acc}_i[k]$ is the acceleration of the i-th train at the k-th communication cycle moment. This state vector contains the train's position, velocity, and acceleration state variables, completely describing the train's motion state.

In the train control system, the velocity and position of the train are regulated by controlling the train's acceleration. Therefore, the acceleration $\mathrm{Acc_i}[k]$ can be regarded as the system's control input, and the train's position $s_i[k]$ and velocity $v_i[k]$ are the controlled statuses of the system.

By establishing the dynamic model of the train, we can analyze the dynamic characteristics of train motion, predict the motion state of the train, and design suitable control strategies according to the control objectives to realize the automatic control of train motion. This is significant for improving the safety, comfort, and efficiency of train operation.

3.2 Problem formulation

The goal of VCTS is to stabilize the distance between adjacent trains and converge each train's speed over time. This implies that VCTS needs to regulate both train speed and acceleration to maintain a constant distance between neighboring trains and eventually synchronize the speeds of all trains.

Mathematically, this goal can be represented as

$$\begin{cases} \lim\limits_{t\to+\infty} \Delta s_i(t) = d(T_i,\ T_{i-1}), \\ \lim\limits_{t\to+\infty} \Delta v_{ij}(t) = 0, \\ \text{Min-ACC} \leqslant \mathrm{Acc_i(t)} \leqslant \text{Max-ACC}, \end{cases}$$

where $\Delta s_i(t) = s_i(t) - s_{i-1}(t)$ and $\Delta v_{ij}(t) = v_i(t) - v_j(t)$. T_i labels the i-th train. $s_i(t)$ and $s_{i-1}(t)$ represent the positions of the i-th and the $(i-1)$-th trains at moment t, and $d(T_i,\ T_{i-1})$ represents the target distance between them. Similarly, $v_i(t)$ and $v_j(t)$ represent the speed of the i-th and j-th trains at time t. $\mathrm{Acc}_i(t)$ refer to the current acceleration of the train, and the maximum acceleration during the acceleration phase, respectively. To maintain passenger comfort, the train's acceleration needs to change slowly within the Minimum Acceleration (Min-ACC) and Maximum Acceleration (Max-ACC) range. This implies that the acceleration cannot change abruptly and must gradually increase or decrease to ensure passenger comfort.

However, traditional VCTS control algorithms could potentially trap agents in local minima instead of the optimal target positions. This means that the conventional control algorithms may converge the train control strategy to a less optimal solution, instead of a globally optimal solution. To avoid local minima, it is essential to modify traditional control machine learning methods and embed them in reinforcement learning.

4 MARL-Based Approach

In this section, we present a novel approach to solving the complex control problem in VCTS using a hybrid reinforcement learning algorithm. Our method, called Hybrid Heuristic-Guided Multi-Agent Proximal Policy Optimization (HHG-MAPPO) (shown in Algorithm 1), combines the power of deep reinforcement learning with heuristic guidance to achieve efficient and safe control of multi-train systems.

4.1 HHG-MAPPO basic environment

State and action representation. The state of the VCTS at any given time k is represented by $S[k] = \{v_{ij}[k],\ x_{ij}[k]\}$, where $v_{ij}[k] = v_i[k] - v_j[k]$ denotes the relative velocity between trains i and j, and $x_{ij}[k] = x_i[k] - x_j[k]$ represents their relative distance. This state representation captures the essential information needed for decision-making in the VCTS.

The action space is discretized into three possible actions: Acceleration ($\mathrm{acc}_0 = \mathrm{acc}^{+}$), maintaining current speed ($\mathrm{acc}_1 = 0$), and deceleration ($\mathrm{acc}_2 = -\mathrm{acc}^{-}$). This discretization allows for precise control while keeping the action space manageable.

Neural network architecture. At the core of our HHG-MAPPO algorithm are two neural networks: A policy network $\pi(a|s;\ \theta)$ and a value network $V(s;\ \phi)$ where θ denotes the policy network parameters, and ϕ denotes the value network parameters. The policy network takes the current state as input and outputs a probability distribution over possible actions, enabling stochastic policy selection. The value network estimates the expected cumulative reward for a given state, which is crucial for computing the advantage function used in policy updates.

Heuristic guidance integration. A key innovation in our approach is the introduction of a heuristic state generator $H(s)$. This component leverages domain knowledge to generate an ideal state s^H based on the current state s. The heuristic guidance is probabilistically incorporated into the action selection process,

Algorithm 1 HHG-MAPPO for VCTS

Require: The state space of the train, $S = \{s_i\}$, $i = 0, 1, \ldots, n$, the action space, $A = \{a_j\}$, $j = 0, 1, \ldots, m$; the reward function, $r = r(s_i, a_j)$; the terminal conditions, S_t; centralized heuristic optimal state generator H; initial heuristic usage rate ε and decay rate α;

Ensure: Policy π and value function V; initialize policy network $\pi(s|\theta)$ and value network $V(s|\phi)$; initialize experience pool D; loss weight λ

for iteration = 1, 2, …, L **do**
  $s_1 \leftarrow$ initialize VCTS state;
  for $t = 1, 2, \ldots, T$ **do** ▷ One episode
    $s_t^H \leftarrow H(s_t)$; ▷ Generate centralized optimal state
    $p \leftarrow$ random(0, 1);
    if $p < \varepsilon$ **then**
      $a_t \sim \pi(s_t^H|\theta)$; ▷ Act based on heuristic state
    else
      $a_t \sim \pi(s_t|\theta)$; ▷ Act based on current state
    end if
    Execute action a_t, observe reward r_t and next state s_{t+1};
    Store transition $\left(s_t, s_t^H, a_t, r_t, s_{t+1}\right)$ in D;
    if $s_{t+1 \in S_t}$ **then**
      Break; ▷ Terminal state reached
    end if
  end for
  for $k = 1, 2, \ldots, K$ **do** ▷ Policy update
    Sample batch of transitions from D;
    Compute advantages A_t using Generalized Advantage Estimation (GAE);
    Compute Proximal Policy Optimization (PPO) loss: $\mathcal{L}_{\text{PPO}} = E_t[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)]$, where E_t denots bath average at stept;
    Compute auxiliary loss: $\mathcal{L}_{\text{aux}} = \frac{1}{n}\sum_{i=1}^{n}\left(V\left(s_t^H|\phi\right) - V(s_t|\phi)\right)^2$;
    Total loss: $\mathcal{L} = \mathcal{L}_{\text{PPO}} - c \cdot E_t[H(\pi(s_t|\theta))] + \lambda \cdot \mathcal{L}_{\text{aux}}$;
    Update θ and ϕ using gradient descent on $\mathcal{L}$;
  end for
  $\varepsilon \leftarrow \varepsilon \cdot \alpha$; ▷ Decrease heuristic usage rate
  Clear experience pool D;
end for
return policy π and value function V

$$A_t = \begin{cases} \sim \pi\left(a|s_t^H;\, \theta\right), & \text{probability } p \\ \sim \pi\left(a|s_t;\, \theta\right), & \text{probability } 1-p \end{cases}.$$

This mechanism allows the algorithm to benefit from expert knowledge, especially in the early stages of learning, while still maintaining the ability to explore and potentially improve upon the heuristic policy.

State transition and reward structure. The state transition follows the dynamics of the VCTS, as shown in Eq. (1),

$$\begin{aligned} v_{k+1} &= v_k + a_k \Delta t, \\ x_{k+1} &= v_k \Delta t + \frac{1}{2} a_k \Delta t^2 \end{aligned} \tag{1}$$

The reward function is designed to encapsulate multiple objectives, as shown in Eq. (2),

$$\begin{aligned} r(s,\, a) = w_1 \cdot r_{\text{distance}} + w_2 \cdot r_{\text{velocity}} + w_3 \cdot r_{\text{acceleration}} + \\ w_4 \cdot r_{\text{safety}} \end{aligned} \tag{2}$$

To improve agents' ability with diverse behaviors/objectives, we consider four methods of creating reward components and combine them into more complex reward functions $r(s, a)$ using the linear function 2. The four reward components is as below. **(1) Weight time distance reward $w_1 \cdot r_{\text{distance}}$:** Encourages agents to maintain an optimal distance from each other. If the distance is too large, a penalty is applied. If a collision occurs, a maximum penalty is applied, ending the training episode. **(2) Weight time velocity reward $w_2 \cdot r_{\text{velocity}}$:** Encourages agents to synchronize their velocities. This helps in maintaining formation and preventing abrupt changes in speed. **(3) Weight time acceleration reward $w_3 \cdot r_{\text{acceleration}}$:** Encourages smooth acceleration. Penalizes sudden changes in acceleration to promote more stable and predictable movements. **(4) Weight time safety reward $w_4 \cdot r_{\text{safety}}$:** Prioritizes overall safety. Penalizes actions that bring agents into dangerous situations, such as moving too close to each other or obstacles. This multi-faceted reward structure encourages the agent to maintain safe distances, synchronize velocities, ensure smooth acceleration, and prioritize overall safety. We note that the combination of these methods allows the creation of complex reward functions with dimensions, enabling the development of agents with diverse behaviors and objectives.

Policy optimization. We employ the PPO algorithm for policy updates, which offers improved stability compared to traditional policy gradient methods. The PPO loss is defined as

$$\mathcal{L}_{\text{PPO}} = E_t[\min(r_t(\theta)A_t,\, \text{clip}(r_t(\theta),\, 1-\epsilon,\, 1+\epsilon)A_t)],$$

where A_t is the advantage function estimated using GAE.

To further leverage the heuristic knowledge, we introduce an auxiliary loss,

$$\mathcal{L}_{\text{aux}} = \frac{1}{n}\sum_{i=1}^{n}\left(V\left(s_t^H|\phi\right) - V(s_t|\phi)\right)^2.$$

This auxiliary loss encourages the value function to align with the heuristic guidance, potentially accelerating learning in critical areas of the state space. The total loss function combines the PPO loss, an entropy term for exploration, and the auxiliary loss,

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} - c \cdot E_t[H(\pi(s_t|\theta))] + \lambda \cdot \mathcal{L}_{\text{aux}},$$

where c is the entropy coefficient, and λ is the weight of the auxiliary loss.

Training process and adaptive parameters. The training process involves iteratively collecting experience, computing advantages, and updating the policy and value networks. To balance the influence of heuristic guidance and learned policy, we introduce heuristic parameters: $\varepsilon \leftarrow \varepsilon \cdot \alpha$, which gradually reduce the reliance on heuristic guidance.

This adaptive approach allows the algorithm to benefit from heuristic knowledge early in training while progressively relying more on learned behavior.

4.2 Handling special cases

To ensure safe and efficient operation, we implement special handling for critical scenarios:

(1) Collision ($x_{ij}[k] \leqslant 0$): Apply high penalty and terminate the episode.

(2) Excessive distance ($x_{ij}[k] > \text{distance}_{\max}$): Apply high penalty and terminate the episode.

(3) Achieving target distance ($x_{ij}[k] = \text{distance}_{ij}$): Provide high reward and terminate the episode.

These special cases help shape the learned policy to prioritize safety and efficiency in VCTS operations. The HHG-MAPPO algorithm represents a significant advancement in VCTS control. By combining the sample efficiency of PPO with heuristic guidance and carefully designed reward structures, we create a learning system capable of rapid adaptation and high-performance control in complex VCTS scenarios. The algorithm's ability to balance heuristic knowledge with learned behavior positions it as a promising approach for real-world VCTS applications, potentially outperforming both traditional control methods and standard reinforcement learning algorithms.

4.3 Description for multi-agent task

At its core, HHG-MAPPO builds upon Policy Optimization, a state-of-the-art reinforcement learning method known for its stability and sample efficiency. The algorithm maintains two neural networks: A policy network $\pi(s|\theta)$ that learns to select actions based on the current state, and a value network $V(s|\phi)$ that estimates the expected cumulative reward for a given state.

A key innovation of HHG-MAPPO is the integration of a centralized heuristic optimal state generator H. This component leverages domain expertise to provide guidance, especially in the early stages of learning. This means that the combination of conventional random selection and experience has effectively reduced and compressed the state space, minimizing the duration of the episode without compromising the accuracy of the train's operational status. The algorithm probabilistically chooses between acting based on the current state or the heuristically generated optimal state, with the probability of using the heuristic guidance decreasing over time. The training process involves iterative episodes where the agent interacts with the VCTS environment. In each time step, the agent selects an action, observes the resulting state and reward, and stores this experience in a memory buffer. The algorithm uses these collected experiences to update the policy and value networks. The policy update process in HHG-MAPPO is particularly sophisticated. It uses the PPO objective, which helps maintain stable learning by limiting the extent of policy changes in each update. Additionally, an auxiliary loss term is introduced to encourage the value function to align with the heuristic guidance. This auxiliary loss helps accelerate learning by leveraging the heuristic knowledge.

To balance exploration and exploitation, the algorithm incorporates an entropy term in its loss function. This encourages the policy to maintain some level of randomness, preventing premature convergence to suboptimal solutions. Throughout training, the algorithm gradually reduces its reliance on heuristic guidance. This is achieved by decreasing both the probability of using the heuristic state for action selection and the weight of the auxiliary loss. This adaptive approach allows the agent to benefit from expert knowledge early in training while progressively relying more on learned behavior, potentially surpassing the performance of the heuristic alone.

The HHG-MAPPO algorithm is designed to address the intricacies of VCTS control. This includes the maintenance of safe distances between trains,

synchronization of velocities, as well as guaranteeing smooth acceleration and deceleration. By integrating reinforcement learning with heuristic guidance, the algorithm strives to provide rapid learning, augmented safety, and, in the end, superior performance in managing VCTS. The key simulation settings and hyperparameters are reported in Table 1.

5 Implementation

The objective of VCTS is to increase transportation efficiency and accommodate passengers. Although the MARL-based controller strives to reduce the disparity between the states of the lead and subsequent trains, the lead train operates at a high velocity. The intention of cooperative control is to ensure that the lead train halts at the appropriate location with a safe level of acceleration, provides passenger comfort during smoother sections of the journey, and has the ability to evade or stop abruptly in unexpected circumstances. However, the stability of dynamic cooperative control could be disrupted. Therefore, in this section, the efficacy of the HHG-MAPPO in the VCTS is evaluated by devising three different scenarios. Further details regarding the specific environmental setup and the results of the experiment will be presented in the succeeding sections.

Table 1 Simulation configurations and parameters. The physical parameters of the train's motion are based on industry standards referenced in Ref. [5], and the reward weight is derived from these standards. The values of ε, α, λ, lr, and p are determined based on preliminary experiments. These values ensure a balanced appropriation of past experiences for learning convergence, supporting both the assistance of past experiences in learning and the rapid convergence of learning. The number of iterations (L), time steps per episode (T), and policy update steps (K) are designed based on the actual operating mechanisms of the train, including the communication frequency between the train and the control center and the computational capabilities.

Parameter	Value
Distance reward weight (w_1)	0.7
Velocity reward weight (w_2)	0.1
Acceleration reward weight (w_3)	0.1
Safety reward weight (w_4)	0.1
Simulation episode (ms)	10–30
Max speed of the train (m/s)	20
Max-Acc (m/s^2)	0.8
Min-Acc (m/s^2)	–0.8
Initial heuristic usage rate (ε)	0.3
Decay rate for heuristic usage rate (α)	0.45
Initial auxiliary loss weight (λ)	0.6
Number of iterations (L)	78–80
Number of time steps per episode (T)	20
Number of policy update steps (K)	10
Batch size (bs)	128
Experience pool size	10 000
Target network update interval (episodes)	200
Learning rate (lr)	0.001
Random probability (p)	0.2

5.1 Multi-agent VCTS environment

In the multi-agent VCTS environment, there are three main scenarios in the trains in reality. The first scenario of "safe acceleration or brake" mentioned refers to a state where the train maintains an appropriate distance, $s_i(t) - s_{i-1}(t)$, between trains under the standards of Max-braking and Max-Acc, and operates at a higher speed safely. In this case, the train can adjust its acceleration or deceleration as needed to ensure safe and efficient operation, but this does not involve restrictive requirements for specific operations such as entering or exiting stations. Such an operational mode allows the train to flexibly respond to various situations during its operation while ensuring safety distance.

In the second scenario of "unpredictable emergency situation", trains operate based on an MA generated by a ZC, where the MA signifies the nearest obstacle approaching the train, which could be another train, a turnout, or a sudden obstruction. The ATP generates a speed limit for the train based on the MA, while the ATO controls the train under the constraints of the ATP. To ensure safety, once the train speed exceeds the limit, an emergency brake must be applied, implying that it is difficult to further reduce the distance between adjacent trains $s_i(t) - s_{i-1}(t)$ due to safety considerations. Thus, disregarding passenger comfort, the distance between adjacent trains must be sufficiently large to prevent collisions.

For the third scenario "uniform schedule operation", this scenario indicates that operations need to be conducted based on the specific requirements of the train's stopping and starting, under the basic condition of "safe acceleration or brake". This means that there are more restrictive conditions when using acceleration, making the operation more complex. For example, when a train is approaching a station, there are explicit speed limitations, and therefore, the train

needs to decelerate in advance to meet the speed requirements at the time of arrival. This operating mode requires the train to follow stricter controls of speed and timing when entering and exiting stations to ensure the safety and punctuality of train operations.

We employ a variety of parameters and hyperparameters to configure and optimize the reinforcement learning algorithm. These parameters and hyperparameters play a crucial role during training, ensuring that the agents can effectively learn and adapt to the environment. Through Eq. (1), it can be observed that the reward components correspond to various aspects such as position, velocity, and acceleration of the train motion system. Velocity is the derivative of position, and acceleration is the derivative related to velocity and system control, with the safety component being positively correlated with both acceleration and velocity. This indicates that our motion system indicators form a chaotic system. Hence, they cannot be analyzed separately. Based on preliminary experimental analysis, it has been concluded that the position is more important than other indicators. We set different reward weights, including distance reward, velocity reward, acceleration reward, and safety reward. Consequently, we have increased the weight of the position reward, with corresponding values of 0.7. And we set different velocity reward, acceleration reward, and safety reward, with corresponding values of 0.1, 0.1, and 0.1. This configuration ensures that the agents not only focus on completing the task quickly but also optimize for safety and stability. The maximum speed of the train is capped at 20 m/s, with Max-Acc and Min-Acc set at 0.8 and −0.8 m/s^2, respectively. These constraints guarantee that agents avoid conducting unstable or unsafe operations stemming from excessively high or low speeds and accelerations. During training, we initialize the heuristic usage rate at 0.3, which is adjusted after each iteration by a decay rate of 0.45. The initial auxiliary loss weight is set at 0.6 and is similarly adjusted over time. The number of iterations is set between 78 and 80, with each iteration consisting of 20 time steps per episode and 10 policy update steps. The batch size is 128, and the experience pool size is 10 000. To maintain the stability and effectiveness of the algorithm, the target network is updated every 200 episodes, and the learning rate is set at 0.001. There is a 0.2 probability of selecting actions based on heuristic states, introducing some randomness to avoid local optima.

These carefully designed parameters and hyperparameters enable effective training of the agents within the multi-agent VCTS environment, ensuring robust performance in complex task settings.

5.2 Simulation result

In our simulation, we examined the functionality of four urban rail train sets, each measuring approximately 120 m, that constitute a VCTS. We constructed three distinct scenarios, each characterized by its unique circumstances and train statuses in Fig. 2.※

In the first scenario, representing safe acceleration or brake, we configured the starting conditions such that the initial distance separating the trains is 50 m, while the initial speed is 0 or 20 m/s. The results showed that the trailing train commenced its movement synchronously with the lead train. Moreover, the trailing train's velocity slightly exceeded that of the lead train. Under this scenario, roughly 84 epochs were required for the state of the VCTS to reach stability. Hence, during safe acceleration or brake, the convergence of rewards exhibited greater smoothness.

In the second scenario, which simulates an unpredictable emergency situation, we specify the starting conditions as follows: The initial distance

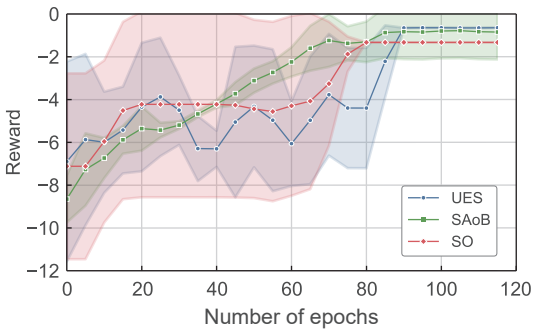


Fig. 2 Performance in terms of reward of the HHG-MAPPO algorithm in three different scenarios: Unpredictable Emergency Situation (UES), Safe Acceleration or Brake (SAoB), and Schedule Operation (SO). Reward changes over training epochs (epoch). The reward function is defined as ($r(s, a) = 0.7 \cdot r_{\text{distance}} + 0.1 \cdot r_{\text{velocity}} + 0.1 \cdot r_{\text{acceleration}} + 0.1 \cdot r_{\text{safety}}$), which evaluates the performance in each scenario. The shaded bands indicate the min–max range across independent runs (random seeds) for each method.

※The shaded bands indicate the min–max range across independent runs (random seeds) for each method. The horizontal axis denotes the training epoch index. Unless otherwise stated, all figures in the simulation result (Section 3.2) follow this plotting convention.

separating each train is set at 10 m, while the initial speed is set at 20 m/s. We observe that in this situation, the reward values tended to fluctuate continuously over time. This is attributable to the requirement for the lead train to execute an emergency evasion maneuver, resulting in abrupt and multiple speed alterations throughout the simulation process. However, by approximately the 92-nd epoch, the reward value in the simulation stabilized and trended toward zero. This implies that the train had successfully performed its emergency evasion and reached its maximum speed of 20 m/s. The leading train is thus able to maintain stability, despite experiencing multiple consecutive speed variations. These observations indicate that the unpredictability inherent in an emergency situation results in fluctuations in rewards among the four trains. Although the speed performance curve of the trailing train exhibited some oscillations, a stable inter-train interaction was restored after a brief adjustment period.

In the third scenario, simulating scheduled operations, the initial inter-train distance is set at 8 m, while the initial speed of all four trains is 20 m/s. Since this configuration adheres to a standard rail timetable, stability is achieved as early as the 20th epoch. Due to the trains' comparatively close proximity, the reward values are not significantly high. Nevertheless, complete convergence is attained within the briefest 80 epochs.

Taken together, these simulations demonstrate prompt convergence in the operational status of the trains across three distinct scenarios, thereby providing substantive verification for our algorithm. This underscores the algorithm's robust short-term adaptability and stability.

Upon carefully examining Figs. 3a and 4a, it becomes quite apparent that following a series of comprehensive tests implemented within the settings of our testing scenario 1, a comparison depicting the variance of acceleration and NRSS is shown through shaded areas and informational tables. This comparison becomes evident when measured against the MAA3C algorithm; our HHG-MAPPO algorithm shows a characteristic of higher stability and winrate.

This stability advantage of our algorithm is particularly crucial in ensuring a consistently high win rate. Furthermore, our algorithm proves competent in

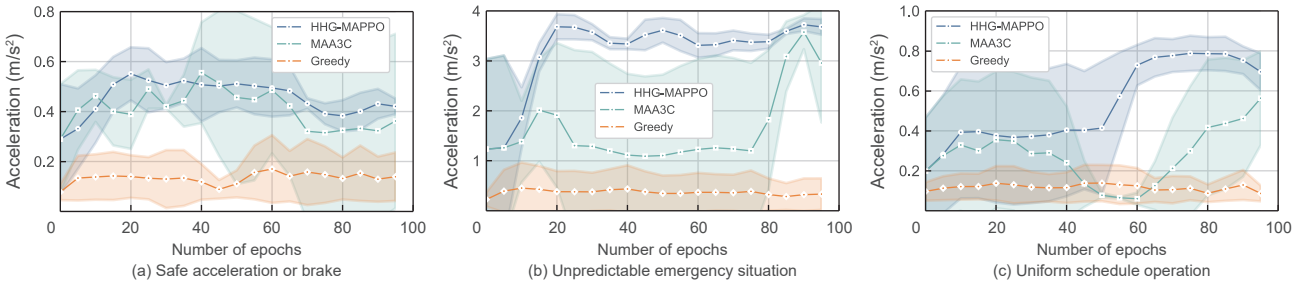


Fig. 3 Performance in terms of acceleration of different algorithms in three types of scenarios. Assuming survival, the smaller the change in acceleration, the better. The shaded bands indicate the min–max range across independent runs (random seeds) for each method.

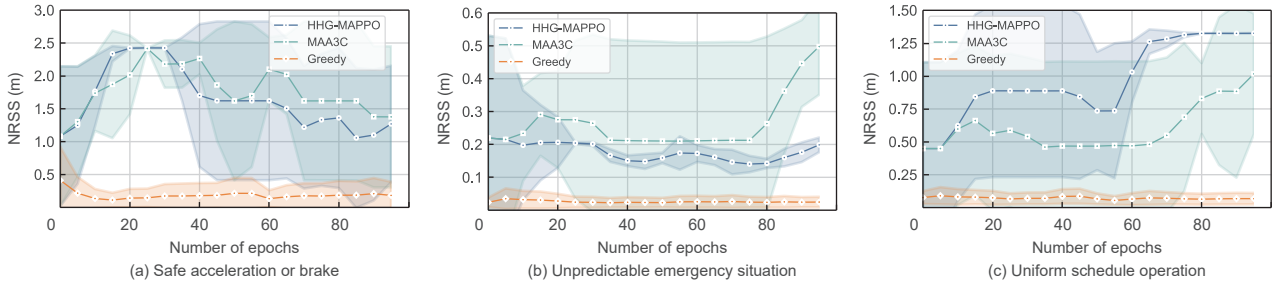


Fig. 4 Normalized Risk Safety Space (NRSS) over training epochs for the HHG-MAPPO, Multi-Agent Asynchronous Advantage Actor-Critic (MAA3C), and Greedy algorithms. NRSS is used to evaluate the deviation of the current vehicle distance from a preset safe distance, ensuring a safe driving distance is maintained. An effective algorithm should aim to maintain a low NRSS throughout the training process. The shaded bands indicate the min–max range across independent runs (random seeds) for each method.

maintaining an optimally safe distance while retaining an acceleration selection process that has been designed for smooth operation. This level of consistency and safety is something that traditional Greedy algorithms have trouble achieving, especially in instances where there is an urgency to meet the multi-agent requirements of the VCTS. One of the unique benefits of our algorithm is also evident from observing the quick stabilization of acceleration changes. Other algorithmic alternatives we assessed either displayed zero growth in acceleration or showcased an amplitude that is simply too high to be considered effective. Both such cases are considered as failing to meet the strict operational requirements that are in place for the effective functioning of VCTS trains.

We further delve into Figs. 3b and 4b which elaborate on how our HHG-MAPPO algorithm is competent enough to respond swiftly in the event of rapid alterations experienced by the leading train which include numerous status shifts as well. Simultaneously, the algorithm works towards achieving effective risk avoidance by retaining a safe distance. Additionally, the MAA3C algorithm falls short when pitted against our algorithm with regard to the win rate and variance across each experimental phase, as depicted clearly in Table 2. This reinforces our initial observation based on the prior scenario, providing further proof that our algorithm shows great concern for stability while ensuring that no collision occurs. This meets, and in many cases exceeds, the operational prerequisites of VCTS trains.

In line with previous findings, Figs. 3c and 4c also show that our algorithm allows for a high win rate and a considerably stable performance even in generalized scenarios that do not involve a pre-defined schedule. This validates the versatility and robustness of our algorithm, further proving its superiority over the existing MARL method and traditional method.

Our study shows that our algorithm quickly adjusts and stabilizes speeding, greatly improving VCTS reliability and efficiency. Ensuring a high win rate and a stable performance, both amidst rapid alterations and in more generalized scenarios, also guarantees an enhanced sense of safety and certainty in urban rail transport. This is evident in Table 2 which shows the consistency in performance under variable conditions.

To address the feedback for a more systematic presentation and detailed comparison with baseline methods, we focus on comparing the results of our HHG-MAPPO algorithm against the MAA3C baseline method. In terms of WinRate, HHG-MAPPO shows an average increment of 23% compared with MAA3C in three scenarios, which underscores our algorithm's enhanced ability to collaborate successfully in the simulated environment. When examining the average indicators, HHG-MAPPO consistently outperforms MAA3C. Specifically, the average NRSS of HHG-MAPPO averages maintain with MAA3C. HHG-MAPPO achieved an average reward in three scenarios

Table 2 Experimental results of WinRate, NRSS, reward, and acceleration for different algorithms in a simulated driving environment. WinRate for collaborative agents is crucial, as a higher WinRate indicates better overall performance and success in collaborative tasks, making the subsequent indicators meaningful (best results are marked in bold). The average of indicator includes NRSS, reward, and acceleration, which provide insights into the efficiency and safety of the algorithms. Lower NRSS values indicate better adherence to safe driving distances, higher rewards reflect more optimal performance, and appropriate acceleration values ensure smooth and safe operations. The variance of these indicators further illustrates the stability and consistency of each algorithm's performance. — means that the method do not use the reward indicator, just use for test for normal basline. Best results are marked in bold.

Algorithm	WinRate for collaborative agent	Average of indicator			Variance of indicator	
		NRSS (m)	Reward	Acceleration (m/s^2)	NRSS (m)	Acceleration (m/s^2)
HHGMAPPO-Scenario-1	**0.75**	1.66	−3.70	0.46	0.76	0.01
MAA3C-Scenario-1	0.70	1.81	−4.36	0.41	0.87	0.04
Greedy-Scenario-1	0.04	0.17	—	0.14	0.03	0.01
HHGMAPPO-Scenario-2	**0.81**	0.17	−4.60	3.32	0.01	0.71
MAA3C-Scenario-2	0.46	0.28	−6.22	1.69	0.05	1.85
Greedy-Scenariso-2	0.00	0.02	—	0.37	0.01	0.06
HHGMAPPO-Scenario-3	**0.75**	1.01	−3.93	0.54	0.21	0.07
MAA3C-Scenario-3	0.45	0.64	−5.89	0.28	0.30	0.10
Greedy-Scenario-3	0.03	0.07	—	0.11	0.01	0.01

of $-(3.7+4.6+3.93)/3=-4.08$ compared with $-(4.36+6.22+5.89)/3=-5.49$ for MAA3C, highlighting a 25.7% increase in achieving optimal performance criteria. In terms of acceleration, HHG-MAPPO maintains smoother and more consistent acceleration figures with a variance of 0.327 m/s^2, in comparison with MAA3C's 0.407 m/s^2. HHG-MAPPO achieved convergence with an average of 87 epochs, while the MAA3C required more than 124 epochs to converge. These detailed comparisons distinctly show the superiority of our HHG-MAPPO algorithm over the baseline MAA3C method in terms of critical performance indicators such as WinRate, NRSS, reward, and stability in acceleration, signifying better performance efficiency and reliability in VCTS.

We believe this innovative algorithm will pave the way for a new era of intelligent and highly efficient urban rail transport systems, surpassing the limitations of traditional methods and existing MARL, while meeting the strict operational requirements for the effective functioning and safety of VCTS trains.

6 Conclusion

This study optimizes co-control strategies for VCTS in urban rail transit using MARL. Advanced technologies boost efficiency and safety to tackle urbanization. We utilize innovative VCTS systems, which bolster co-control between trains by facilitating efficient vehicle-to-vehicle interaction. The study uses MARL, ensuring effective coordination amid multiple agents. We validate our MARL algorithm via simulations, showing its superiority over traditional methods. We plan to fine-tune these algorithms and test them in real urban rail systems. Despite the promising results obtained using MARL for optimizing co-control strategies in VCTS for urban rail transit, challenges remain. One limitation is the scalability of the MARL algorithm when applied to more complex and larger-scale networks. Furthermore, our algorithm's reliance on high-quality, real-time data may restrict its use in environments where such data inputs are inconsistent or unavailable. In future work, we will expand this approach to other IIoT applications like autonomous driving, greatly contributing to urban mobility and improving passenger experience.

Acknowledgment

This work was in part supported by the Major Basic Research Program of Shandong Provincial Natural Science Foundation (No. ZR2022ZD02), the Joint Key Funds of National Natural Science Foundation of China (No. U24A20244), the National Natural Science Foundation of China (No. 62272256), and Shandong Science Fund for Excellent Young Scholars (No. 2023HWYQ-007).

Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

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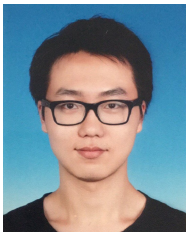
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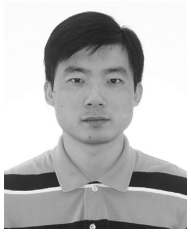
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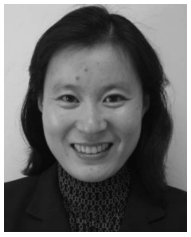
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